

Scalable Multi-Tier AI Data Ingestion Architecture

1. Scale Strategy (500,000+ Records):

- **Ingestion Layer:** Distributed asynchronous scraping architecture built using asyncio and Playwright/aiohttp for non-blocking I/O.
- **Queue System:** Celery with Redis for task distribution across multiple scraper nodes.

2. Handling Rate Limits (429) & Context Window Limits (413):

- **Fallback Strategy:** LiteLLM Orchestrator dynamically routes requests: Gemini Flash $\rightarrow$ Groq Llama 3 $\rightarrow$ DeepSeek.
- **Backoff & Chunking:** Exponential backoff with jitter for HTTP 429s. Text payloads are dynamically chunked/truncated using semantic sentence splitting to avoid 413 context overflows.

3. Data Freshness & Deduplication:

- SHA-256 Hashing on normalized target URLs stored in Redis to guarantee content is ingested only once within a 24-hour window.

4. Storage Architecture:

- **Primary DB:** PostgreSQL for structured entities (Startups, Products, Papers).
- **Graph / Vector Storage:** Neo4j / Pinecone for mapping complex startup-to-product relationships and semantic deduplication.